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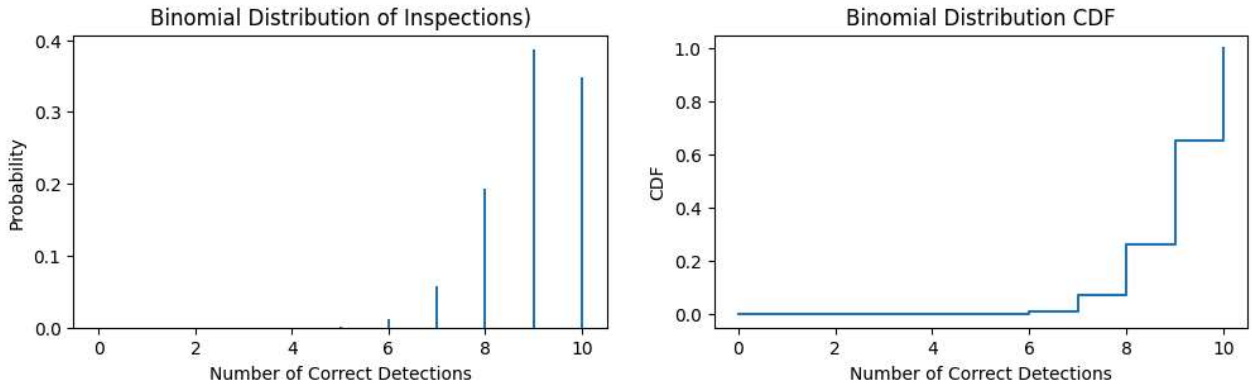
```
import numpy as np
import matplotlib.pyplot as plt
import random
import math
```

```
# Binomial parameters
n = 10        # number of components inspected
p = 0.9       # probability of correct detection

# Random variable X
X_b = np.arange(0, n + 1)    #X_b: Binomial random variable X

# Binomial PMF
binomial_pmf = []
for x in X_b:
    prob = math.comb(n, x) * (p ** x) * ((1 - p) ** (n - x))
    binomial_pmf.append(prob)

# Binomial CDF
binomial_cdf = np.cumsum(binomial_pmf)
```

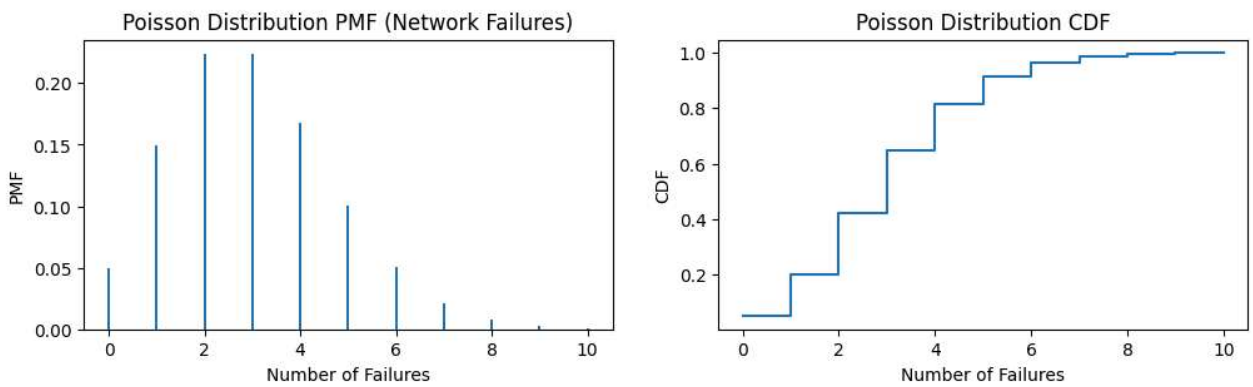
```
# Binomial CDF
binomial_cdf = np.cumsum(binomial_pmf)

#Plotting pmf and cdf
plt.figure(figsize=[12,3]) #size of plot in inches

plt.subplot(121) #first subplot
plt.bar(X_b, binomial_pmf,width=0.05)
plt.xlabel("Number of Correct Detections")
plt.ylabel("Probability")
plt.title("Binomial Distribution of Inspections)")

plt.subplot(122) #Second subplot
plt.step(X_b, binomial_cdf, where='post')
plt.xlabel("Number of Correct Detections")
plt.ylabel("CDF")
plt.title("Binomial Distribution CDF")
plt.show()
```




```
# Binomial CDF
binomial_cdf = np.cumsum(binomial_pmf)
# Simulation: Taking 10000 batch of sample size 10 and finding nmber of corrrect detection.
binomial_sim = np.random.binomial(n, p, 10000)

# Event: Exactly 8 correct detections
sim_prob_8 = np.mean(binomial_sim == 8)
theo_prob_8 = binomial_pmf[8]

print("BINOMIAL DISTRIBUTION COMPARISON FOR P(X=8)")
print(f"Theoretical Probability:{theo_prob_8:0.4f}")
print(f"Simulated Probability:{sim_prob_8:0.4f}")
```

```
BINOMIAL DISTRIBUTION COMPARISON FOR P(X=8)
Theoretical Probability:0.1937
Simulated Probability:0.1932
```

```
#Poission Distribution

import numpy as np
import matplotlib.pyplot as plt
import math

# Parameter
lam = 3         # average failures per hour
X_p = np.arange(0, 11) #Taking values of RV till X=10

# Poisson PMF
poisson_pmf = []
for x in X_p:
    prob = (math.exp(-lam) * (lam**x)) / math.factorial(x)
    poisson_pmf.append(prob)

# Poisson CDF
poisson_cdf = np.cumsum(poisson_pmf)

#Plotting pmf and cdf
plt.figure(figsize=[12,3]) #sise of plot in inches

plt.subplot(121) #first subplot poission pmf
plt.bar(X_p, poisson_pmf,width=0.05)
plt.xlabel("Number of Failures")
plt.ylabel("PMF")
plt.title("Poisson Distribution PMF (Network Failures)")

plt.subplot(122) #Second subplot Poisson CDF
plt.step(X_p, poisson_cdf, where='post')
plt.xlabel("Number of Failures")
plt.ylabel("CDF")
plt.title("Poisson Distribution CDF")
plt.show()
```




```
# Simulation: Finding number of failures in 10000 samples
poisson_sim = np.random.poisson(lam,10000)

# Event: At most 2 failures
sim_prob_le_2 = np.mean(poisson_sim <= 2)
theo_prob_le_2 = sum(poisson_pmf[:3])

print("\nPOISSON DISTRIBUTION COMPARISON")
print("P(X ≤ 2)")
print(f"Theoretical Probability:{theo_prob_le_2:0.4f}")
print(f"Simulated Probability:{sim_prob_le_2:0.4f}")
```

```
POISSON DISTRIBUTION COMPARISON
P(X ≤ 2)
Theoretical Probability:0.4232
Simulated Probability:0.4383
```